Korea Technology and Information Promotion Agency for SMEs

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	***AI Foundation Models*** *for Warpage in* ***Advanced Packaging*** *to Secure a Technological Lead in* ***High Bandwidth Memory*** *Manufacturing*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), AI (√), Advanced Robotics & Manufacturing (√)			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	2. Electrical & Electronics		*7. Large Classification: Knowledge Service*	
	Medium Category	Small Category	Medium Category	Small Category
	Semiconductor Device and System (2004)	*Design Tool (200407)*	*Manufacturing Engineering Service (7005)*	*Production Process Modeling/Simulation Service (700503)*
Concept of Proposed Technology	***Core concept.*** *Korea occupies a pivotal position in the global semiconductor supply chain, particularly in HBM, and strengthening the growth and technological competitiveness of Korea's 소·부·장 (MPE: materials, parts, and equipment) companies is a critical priority for the national semiconductor ecosystem. As the industry paradigm rapidly shifts toward Advanced Packaging and Heterogeneous Integration, the domestic MPE ecosystem is also required to undergo a corresponding technology transition. In response, Korean semiconductor SMEs are actively focusing*			

	on securing core technologies aligned with this new paradigm. *In particular, in 3D stacking processes—including HBM—warpage has emerged as a key quality factor that directly affects yield and reliability. Accordingly, the importance of manufacturing technologies as well as Metrology & Inspection (M&I) capabilities that can precisely measure, analyze, and control warpage continues to grow. In fact, many Korean SMEs are moving to enter the M&I equipment market for HBM processes and to supply related measurement and analysis solutions. (For example, Hanmi Semiconductor achieved exceptional growth by securing stack equipment technology, demonstrating that acquiring core equipment technologies can serve as a catalyst for corporate growth.)* *Hence, collaborative research targeting warpage prediction, control, and metrology technologies that can be immediately applied in Advanced Packaging manufacturing will provide a technical foundation that Korean equipment companies can rapidly translate into practical business models, ultimately contributing to enhanced competitiveness and accelerated growth of Korea's MPE ecosystem.* *As wafer (die-level) warpage measurements and computer simulation outputs become more readily available, this trend presents an opportunity to construct AI Foundation Models that can learn broad patterns and representations (of wafer warpage) from vast amounts of heterogeneous data, and then serves as a foundation for various downstream applications via fine-tuning on additional data (given specific focuses or needs from one or even multiple Korean SMEs).* ***Main function and potential uses**:* • *Model and predict die-level wafer warpage at given conditions* • *Enable new strategies to control wafer warpage* • *Optimization for wafer 3D stacking* ***Some potential (but are not limited to) Korean SMEs partners:*** • *ATI (Advanced Technology Inc.)* • *Nexensor* • *Kovis* • *Auros* • *Pemtron* • *Cressem* • *Ohlabs* • *Acoulab*

Rationale for Joint R&D Project	***Rationale for collaboration.*** *The nature of AI Foundation Models requires a close collaboration between Georgia Tech and an industrial partner:* • ***The Georgia Tech team*** *provides the theoretical expertise of process quality, stream of variation, AI and machine learning (needed for constructing the Foundation Model), while* ***the Korean SME partner*** *provides the downstream applications, essential domain knowledge, data and testbed that enable the fine-tuning of the Foundation Model.* ***Expected benefits of joint R&D.*** • *Without the complementary expertise, data and resources from both partners, building the AI foundation model for wafer warpage is not possible.* • *Accelerate the transition from model development, testing to implementation.* • *Reduced technical risk through iterative algorithm–hardware co-development and validation.*
Key Words	*The key terms that describe the proposed technology and its main features include:* 1. *AI Foundation Models* 2. *Wafer Warpage* 3. *3D Stacking* 4. *High Bandwidth Memory (HBM)* 5. *Metrology and Inspection (MI)*

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]		**Start of the Project**	*Level (2)*	**End of the Project**	*Level (6)*
R&D Objectives		*The specific technologies to be developed through this R&D project:* • *To develop a physics-informed generative AI tool for chip-level high-precision wafer warpage, which not only integrates data from the high-resolution 3D wafer warpage measurement and numerical simulations using the finite element methods, but also provides the capability to generate wafer warpage scenarios as a generative AI tool for design and process optimization.* • *To develop, together with Korean SMEs, a non-uniform film formation technology for warpage compensation of wafers (or chips) leveraging the high-precision wafer modeling capabilities developed above.* • *To reduce the cost of high-resolution 3D wafer warpage data by developing an in-wafer virtual metrology methodology to reconstruct full-wafer data from partial in-wafer sampling.*			

R&D Target Specifications		**[Task 1. Physics-informed generative AI modeling and predictions]** *1.1. Develop geometric representation method (HBM-ready)* • *Specification: Develop an effective mathematical/geometric representation for warpage and establish a "Chips-on-Wafer" geometric model suitable for HBM products and processes. (Y/N)* • *Target level: Yes* • *Remarks: Deliverable check* • *Description: Define the warpage geometry of chips and wafers (e.g., representation form, parameters, modes) and finalize it as a model structure that can be used for subsequent learning and prediction.* *1.2. High-precision warpage behavior prediction (chip/die level)* • *Specification: Correlation coefficient r ≥ 0.90 between predicted and measured warpage, and root mean squared error (RMSE) at the die level.* • *Target level: r ≥ 0.90, RMSE ≤ 5 μm.* • *Baseline/remarks: Exceeds the performance reported in prior SK hynix Inc. results [2], where large-scale chip-level warpage measurement and validation achieved r ≥ 0.66.* • *Description: Develop a physics-informed generative AI model that predicts wafer and individual die warpage behavior across key process steps (e.g., wafer thinning, film formation, sawing/dicing, thermal processes). Validate prediction accuracy using large-volume experimental data and quantify model performance via correlation.* **[Task 2. Wafer warpage control and variation reduction]** *2.1. Develop a warpage control method* • *Target specification: Develop a warpage control methodology.* • *Target level: Yes (deliverable completed).* • *Remarks: Must reflect process and equipment constraints.* • *Description: Develop a warpage compensation approach (e.g., non-uniform film formation) along with a control algorithm/recipe and demonstrate feasibility for practical deployment.* *2.2. Warpage removal (flattening) performance* • *Target specification: PV (peak-to-valley) reduction rate ≥ 80% compared to the pre-control baseline.* • *Target level: ≥ 80%.* • *Remarks: PV is a primary "before/after" warpage magnitude metric.* • *Description: Quantify how much warpage is removed by comparing PV*

		(max–min out-of-plane displacement) before and after control. *2.3. Global curvature uniformity* • *Target specification: Improvement rate of global curvature standard deviation (σ_κ) ≥ 80% (alternatively, directly target an absolute threshold such as σ_κ ≤ target value).* • *Target level: ≥ 80%.* • *Remarks: Estimate curvature by region (grid/die-level), then evaluate σ_κ .* • *Description: Evaluate how uniformly global curvature is distributed across the wafer after control by measuring the reduction in curvature dispersion (standard deviation).* **[Task 3. in-wafer Virtual Metrology]** *3.1 Develop in-wafer Virtual Metrology* • *Target specification: Develop an in-wafer Virtual Metrology (VM) methodology. (Y/N)* • *Target level: Yes* • *Remarks: Deliverable check* • *Description: Establish a VM algorithm/workflow that reconstructs full-wafer high-resolution warpage (or inspection characteristics) from partial in-wafer sampling data.* *3.2 Measurement point reduction rate* • *Target specification: Reduce the number of measurement points by ≥ 50% compared to full-scan measurement, under the same accuracy constraint (within allowable error).* • *Target level: ≥ 50%.* • *Remarks: Compare under identical accuracy/allowable error requirements.* • *Description: Reduce measurement burden (time/cost) by minimizing sampling points while maintaining reconstruction accuracy.* *3.3 Throughput improvement* • *Specification: improvement in inspection time (APW: average per wafer).* • *Target level: 50%* • *Remarks: Standard conditions must be specified (e.g., 50 MHz transducer, single transducer, 300 mm full-wafer SAT inspection conditions).*

		• *Description: Improve equipment throughput by reducing inspection time. For example, reducing time per wafer from 20 minutes to 10 minutes corresponds to a 50%-time reduction; in wafer/hour terms, this can be interpreted as ~2× throughput (+100%), depending on the definition used.*
R&D Plan		*The detailed plan of the research approach, key tasks, methodologies, expected milestones, and the timeline required to achieve the stated R&D objectives* *The proposed R&D plan consists of three major tasks: physics-informed generative AI modeling and predictions (Task 1), Wafer warpage control and variation reduction (Task 2), and in-wafer virtual metrology (Task 3).* ***Task 1: physics-informed generative AI modeling and predictions*** *In Task 1, the primary goal is to develop a generative diffusion model that is able to generate high-precision chip-level wafer warpage utilizing both 3D wafer warpage measurements and low-fidelity numerical simulation outputs. In particular, Task 1 will address two questions: (1) What is the most effective mathematical geometric representation for wafer warpage using generative diffusion models? and (2) How can the physics-informed generative AI model be trained and how can the data be generated?* *The completion of Task 1 will enable the key capability of predicting high-resolution wafer warpage by running the low-fidelity model under various design or process parameter settings, and then using a pretrained diffusion model to re-construct the high-resolution output.* ***Task 2: wafer warpage control and variation reduction*** *In Task 2, the primary goal is to develop a non-uniform film formation technology for warpage compensation of wafers (or chips). Current practical approaches to warpage control mainly rely on uniform thin film deposition, which is constrained by Stoney's equation and thus limited in its ability to achieve precise compensation. Most methods remain at the level of experimental or empirical adjustments, while mathematical model–based precision control frameworks are still at an early stage. In particular, Task 2 will provide solutions to the following questions: (1) What is the optimal non-uniform film profile tailored to each individual wafer to alleviate warpage? This approach enables the induction of non-uniform stress distributions across local regions, thereby overcoming the inherent limitations of conventional uniform thin film deposition methods. (2) How to perform variation reduction of the controlled wafer warpage? A stochastic optimization problem with a chance constraint can be formulated to find the most robust wafer control solution (i.e., non-uniform film profile) that achieves desirable variation*

		reduction. ***Task 3: in-wafer virtual metrology*** *In Task 3, the primary goal is to develop an in-wafer Virtual Metrology (VM) methodology via cost-effective data sampling to reconstruct full-wafer data from in-wafer sampling and propose an optimal sampling strategy. In particular, Task 3 will address the following questions: (1) How to reconstruct the high-resolution wafer warpage data from sparse measurement data? (2) How to determine the cost-effective sampling strategy, i.e., locations to be measured? The capability to reconstruct the high-resolution wafer warpage above enables us to formulate the optimal sampling problem into a framework of compressed sensing, for which the goal is to find the sparsest measurement locations, subject to the physical constraints and characteristics of actual measurement & inspection technologies and equipment, that yield the high-resolution wafer warpage within the limit of the maximum error allowed.* ***Timelines.*** *This is a 3-year R&D project, and the proposed timelines are given as follows:* • *Quarter 1-3: A Bézier-surfaces-based Random-Field mathematical model that represents chip-level high-resolution wafer warpage by random tensors. It provides a tensor-based high-resolution geometric representation, uncertainty quantification for wafer-to-wafer variation, and the foundation for generative diffusion models for wafer warpage.* • *Quarter 2-6: A DDPM generative diffusion model for wafer warpage, and a reconstruction algorithm for generating high-resolution wafer warpage using low-fidelity numerical simulations.* • *Quarter 3-8: A wafer warpage control strategy through non-uniform thin film deposition and an algorithm to determine the optimal non-uniform film profile. A variation reduction technique, based on a tensor-to-tensor regression, to control the variation of the after-control wafer warpage.* • *Quarter 5-8: A reconstruction algorithm for generating high-resolution wafer warpage using sparse wafer warpage measurements. D6). A cost-effective sampling strategy that reduces the time and cost by taking sparse wafer warpage measurements in a framework of compressed sensing.* • *Quarter 9-12: Testing, implementation and improvement.*

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Describe the expected roles and responsibilities of the research institute, such as technical contributions, research scope, resources to be provided, and specific tasks to be provided, etc.* *The Georgia Tech team will be responsible for the development of core AI algorithms (and provide expertise in process quality, stream of variation, AI and machine learning) including* • *Develop the (1) AI Foundation Models (a diffusion model) in Task 1, (2) mathematical formulation and solution of the optimization problem for wafer warpage control in Task 2, and (3) cost-effective data sampling algorithm in Task 3.* • *Support the model testing, validation and implementation, together with the Korean SME partners.* • *Provide access to latest theoretical advancement and world-class faculty/graduate students* • *Participate in regular meetings and discussions with the Korean SME partners.* • *Documentation of the developed technologies.*
Expected R&R of Korean SME Partner	*Describe the expected roles and responsibilities of Korean SME, such as technical contributions, development tasks, testing/validation activities, resource commitments, and collaborations efforts, etc.* *The Korean SME partner provides the essential domain knowledge, data and experimental/validation platform, and guidance and relevance to real challenges. For example, FEM-based or simplified simulation results for warpage behavior (if any), full-wafer scan data set for training and validation data shared in anonymized/normalized form, discussions on current deposition process control, identifying and utilizing available metrology processes, real testing use cases, etc.*

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*Explain the expected outcomes of this joint R&D project, including technological achievements, commercialization potential, and broader industrial or economic impact, etc.* *The partnership will accelerate the penetration of the latest development of AI into semiconductor design and manufacturing, and strengthen the leadership position of Korean SMEs in this field. The collaboration is expected to generate a list of potential outcomes and impact. For example,* • *A Bézier-surfaces-based Random-Field mathematical model that represents chip-level high-resolution wafer warpage by random tensors. It provides a tensor-based high-resolution geometric representation, uncertainty quantification for wafer-to-wafer variation, and the foundation for generative diffusion models for wafer warpage.* • *A DDPM generative diffusion model for wafer warpage, and a reconstruction algorithm for generating high-resolution wafer warpage using low-fidelity numerical simulations.* • *A wafer warpage control strategy through non-uniform thin film deposition and an algorithm*

	to determine the optimal non-uniform film profile. • *A variation reduction technique, based on a tensor-to-tensor regression, to control the variation of the after-control wafer warpage.* • *A reconstruction algorithm for generating high-resolution wafer warpage using sparse wafer warpage measurements.* • *A cost-effective sampling strategy that reduces the time and cost by taking sparse wafer warpage measurements in a framework of compressed sensing.* • *Software: the methodologies/algorithms above will be delivered in the form of software packages.* • *Reports: the team will deliver annual progress reports, as well as a final testing and validation report at the end of the project.*

6. References

(1) Lee, S., Liu, X. and Shi, J., 2026. A Gaussian Process for Chips on Wafer Warpage Modeling based on Bezier Surface. *Working Paper*.

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(4) Yoon, S.H., Lee, S.Y. "Enhancing HBM3E KGSD (Known Good Stacked Dies) Quality Competitiveness Through the Implementation of Gray AI-Based Technology," The 32nd Korean Conference on Semiconductors, Jeongseon, Korea, 2025.

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(8) Assignee, Wafer shape and flatness measurement apparatus and method, U.S. Patent 11,105,753 B2, Sep. 7, 2021. https://patents.google.com/patent/US11105753B2/en

(9) Assignee, Wafer shape and thickness measurement system utilizing shearing interferometers, U.S. Patent Application US 20,140,293,291 A1, Oct. 2, 2014. https://patents. google.com/patent/US20140293291A1/en

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization